

**COPYRIGHT RESERVED BCA (Sem-I) – Programming
Fundamental Using C++ (101) Core-I**

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

Compulsory

1. Choose the correct alternative for each of the
following questions : 2×10=20

(a) Which of the following approach is used
by C++ ?

- (i) Left-right
 - (ii) Right-left
 - ☒ (iii) Bottomup
 - (iv) Top down
- (b) Which of the following correctly declares an array in C++ ?
- (i) array{10}
 - ☒ (ii) array array [10]
 - (iii) int array;
 - (iv) int array {10};
- (c) Which is more effective while calling the C++ function ?
- ☒ (i) Call by object
 - (ii) Call by pointer
 - ☒ (iii) Call by value
 - (iv) All of the above
- (d) What is inheritance in C++ ?

- ☒ (i) Deriving new class from existing class
 - (ii) Overloading of classes
 - (iii) Classes with same name
 - (iv) Wrapping of data into a single class
- (e) The if else statement can be replaced by which operator :
- (i) Bitwise operator
 - ☒ (ii) Conditional operator
 - (iii) Multiplicative operator
 - (iv) Addition operator
- (f) The switch statement is also called as
- ☒ (i) Choosing structure
 - (ii) Selective structure
 - (iii) Certain structure
 - ☒ (iv) Bitwise structure
- (g) The destination statement for the goto lable is identified by what lable ?

(i) \$

(ii) @

(iii) *

~~(iv)~~ :

(h) What type of comment does C++ support ?

(i) Single line

(ii) Multiline

~~(iii)~~ Single line and Multiline

(iv) Resuable line

(i) Where does the execution of the program starts :

(i) User defined function

~~(ii)~~ Main function

(iii) Void function

(iv) Else function

(j) An inline function is expanded during —

~~Compile~~ time

Runtime

Never expanded

End of the program

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

2. Write down the difference between OOP and POP.

~~3.~~ Define constructor and briefly explain its type.

4. Explain overloading vs overriding.

~~5.~~ Write down the features of object oriented programming.

~~6.~~ What are the different types of loop in C++ ?

7. Write down any three string built in function with explanation.

8. Explain the concept of encapsulation with example.

9. What do you mean by polymorphism ? Explain with help of example.

Section-C

Long Answer type Questions

Answer any **three** questions of the following :

10×3=30

10. Write a program in C++ that gives the user the option of converting fahrenheit to celsius or celsius to fahrenheit and depending upon user's choice carries out the conversion.
11. Write a program in C++ to input a character and change its case to Upper case if the character is entered in lower case and vice versa.
12. What is Inheritance ? Explain it types with proper example.
13. Define a class candidate in C++ with the following description :

Private member

A data member RNO of type long

A data member scare of type float

A data member Name of type string

A data member Remarks of type string

A member function Assign Rem () to assign remarks as per the scare

Scare	Remark
>=50	Selected
less than 50	Not selected

Public members

A function ENTER () to allow user to enter value for RNO, Name, Scare and call function Assign Rem () to assign Remarks.

A function DISPLAY () to allow user to view the content of all the data members.

14. Write short notes on :

- (a) Data Abstraction**
- (b) Pointer**
- (c) Array**
- (d) Destructor**

15. Write a program in C++ to find the factorial of a given number.

_____ x _____